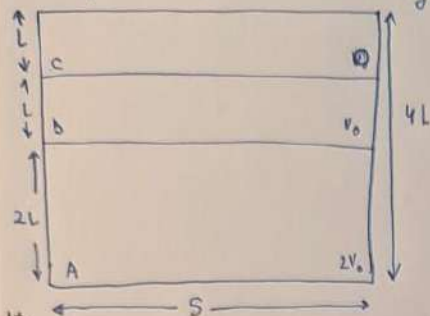


5.3 A classical gas at equilibrium contains N non-interacting molecules that move inside a container with cross-sectional area S and height $4L$ as shown in Fig 5.9. A field $V=V(x)$ acts on the gas molecules as indicated. Additionally, each particle is characterised by internal degrees of freedom such that it can stay in 2 quantum energy states, 0 and ϵ with degeneracy 1 and 3, respectively



$\epsilon_1 = 0 \quad \epsilon_2 = \epsilon$
 $g_1 = 1 \quad g_2 = 3$

$$V(x) = \begin{cases} 0 & 0 \leq x \leq 2L \\ V_0 & 2L \leq x \leq 3L \\ 2V_0 & 3L \leq x \leq 4L \end{cases}$$

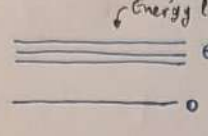
a) Calculate the canonical partition function of this system

$$Z = \frac{1}{N!} (Z_1)^N \quad \text{with } Z_1 = Z_{1,cl} \cdot Z_{1,int}$$

Indistinguishable

$$Z_{1,int} = \sum_i g_i e^{-\beta \epsilon_i} = 1 \cdot e^{-\beta \cdot 0} + 3e^{-\beta \cdot \epsilon} = 1 + 3e^{-\beta \epsilon}$$

Energy levels



$$Z_{1,cl} = \frac{1}{h^3} \int e^{-\beta H_1} d\vec{q} d\vec{p} = \frac{1}{h^3} \int dq_x dq_y \int_0^{4L} dq_z e^{-\beta \left(\frac{p_x^2}{2m} + \frac{p_y^2}{2m} + \frac{p_z^2}{2m} + V(z) \right)}$$

Isotropic

$$Z_{1,cl} = \frac{S}{h^3} \left[\int_0^{2L} e^{-\beta 2V_0} dq_z + \int_{2L}^{3L} e^{-\beta V_0} dq_z + \int_{3L}^{4L} e^{-\beta \cdot 0} dq_z \right] \left[\int_{\mathbb{R}} e^{-\beta \frac{p^2}{2m}} d\vec{p} \right]^3$$

$$Z_{1,cl} = \frac{S}{h^3} \left[2Le^{-2\beta V_0} + Le^{-\beta V_0} + L \right] \left[\int_{\mathbb{R}} e^{-\frac{p^2}{2mKT}} d\vec{p} \right] = \frac{SL}{h^3} \left[2e^{-2\beta V_0} + e^{-\beta V_0} + 1 \right] \left[\sqrt{\frac{2\pi mKT}{h^2}} \right]^3$$

Variable change

$$SL = \frac{1}{4} V$$

Gaussian integral = $\sqrt{\pi}$

$$Z_{1,cl} = \frac{V}{4h^3} (2e^{-2\beta V_0} + e^{-\beta V_0} + 1) (2\pi mKT)^{3/2}$$

De Broglie thermal wavelength

$$\Lambda = \frac{h}{(2\pi mKT)^{1/2}}$$

$$Z_{1,cl} = \frac{V}{4\Lambda^3} (2e^{-2\beta V_0} + e^{-\beta V_0} + 1)$$

For a monoatomic ideal gas with no field applied

$$Z_{1,cl} = \frac{V}{\Lambda^3}$$

$$Z_1 = (1 + 3e^{-\beta \epsilon}) \frac{V}{4\Lambda^3} (2e^{-2\beta V_0} + e^{-\beta V_0} + 1) \Rightarrow Z = \frac{1}{N!} (1 + 3e^{-\beta \epsilon})^N \frac{V^N}{4^N \Lambda^{3N}} (2e^{-2\beta V_0} + e^{-\beta V_0} + 1)^N$$

$$P_A = \frac{\int_A d\vec{q} d\vec{p} e^{-\beta H}}{\int d\vec{q} d\vec{p} e^{-\beta H}} = \frac{2e^{-\beta 2V_0}}{2e^{-\beta 2V_0} + e^{-\beta V_0} + 1}$$

$$P_B = \frac{e^{-\beta V_0}}{Z_{1,cl}} \quad P_C = \frac{1}{Z_{1,cl}}$$

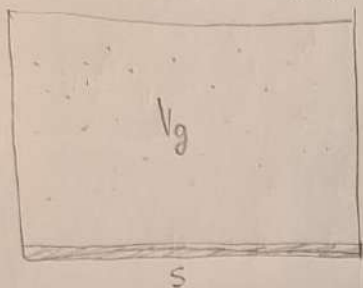
Classical partition function

	$T \rightarrow 0$	$T \rightarrow \infty$
P_A	0	1/2
P_B	0	1/4
P_C	1	1/4

$$P(C|B) = P_C P_B = \frac{3e^{-\beta \epsilon}}{Z_{1,int}} \frac{e^{-\beta V_0}}{Z_{1,cl}}$$

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N ideal non-interacting gas particles of mass m at thermal eq., T



$$E = \frac{p^2}{2m} - \epsilon_0$$

↑
kinetic energy

↓
binding energy

solid surface

$$Z_g = \frac{1}{N_g} (Z_{1,g})^{N_g}$$

$$Z_s = \frac{1}{N_s} (Z_{1,s})^{N_s}$$

$$N = N_g + N_s$$

$$Z_{1,g} = \frac{1}{h^3} \int_{\mathbb{R}^3} d^3q d^3p e^{-\beta \frac{p^2}{2m}} = \frac{V_g}{h^3} (2\pi m kT)^{3/2} \cdot (\pi)^3 = \frac{V_g}{\Lambda^3}$$

3 = log 2

$$\left(u = \frac{p}{\sqrt{2m kT}} \right) \quad \left(du = \frac{dp}{\sqrt{2m kT}} \right)$$

↑
Gaussian int.

$$Z_{1,s} = \frac{1}{h^2} \int_{\mathbb{R}^2} d^2q d^2p e^{-\beta \left(\frac{p^2}{2m} - \epsilon_0 \right)} = \frac{S}{h^2} e^{+\beta \epsilon_0} 2\pi m kT = \frac{S}{\Lambda^2} e^{\beta \epsilon_0}$$

$$G = A + PV = -kT \ln Z + NkT$$